

Supplementary Material: Repair-Aware Quantum Sampling for a Discrete Process-Design QUBO

Yirang Park^a and David E. Bernal Neira^a

^a Davidson School of Chemical Engineering, Purdue University, West Lafayette, IN 47907, USA

ABSTRACT

This supplement records provenance, validation, and artifact anchors for the DS-MFG repair-aware quantum-sampling manuscript. It is intended to be compiled and submitted separately from the main conference paper if appendices are not counted as article body material in the submission workflow.

Keywords: supplementary material, quantum optimization, process design, reproducibility

REPOSITORY AND PACKAGING

The main manuscript keeps only the narrative-critical values. This supplement keeps the audit data close to the manuscript source while allowing the main article to remain within the conference page limit. The repository URL is <https://github.com/SECQUOIA/pse-quantum-discrete-optimization>. A final archival DOI has not yet been assigned. It will be added before camera-ready submission if one is minted; the review artifact is identified by the repository snapshot that contains this supplement and by the checksums below.

Table 1: Current reproducibility pins validated for the DS-MFG case study.

Layer	Current pin or check	Repository anchor
Julia optimization wrapper	QiskitOpt.jl v0.7.0; manifest git tree a601dc cdd43fd0a8b18e63f3b771273817af49eb.	case-studies/ds-mfg-qubo-qiskitopt/Project.toml; case-studies/ds-mfg-qubo-qiskitopt/Manifest.toml.
Julia QUBO stack	QUBODrivers.jl v0.6.1, QUBOTools.jl v0.13.0, QUBO.jl v0.6.0.	case-studies/ds-mfg-qubo-qiskitopt/Project.toml; case-studies/ds-mfg-qubo-qiskitopt/Manifest.toml.
Python bridge	Qiskit 2.3.1, Qiskit Aer 0.17.2, Qiskit Optimization 0.7.0, Qiskit IBM Runtime 0.46.1, NumPy 2.2.6, SciPy 1.15.3.	case-studies/ds-mfg-qubo-qiskitopt/CondaPkg.toml.
Fresh-clone validation	julia --project=. scripts/smoke_test.jl imports packages, checks required artifacts, and validates cached summary values.	case-studies/ds-mfg-qubo-qiskitopt/scripts/smoke_test.jl.

Table 2: Reproducibility levels used for the DS-MFG manuscript package.

Level	Meaning	Current support
Score-from-cache reproducible	Retained samples, summaries, and repair tables can be inspected and rescored without rerunning the stochastic campaigns.	Supported by cached distributions, hit-rate reports, checksums, and the smoke test. This is the main reproducibility level for the manuscript claims.
Rerunnable with retained scripts	The scripts and pinned environment can rerun selected QAOA/VQE, classical, and hardware-handoff workflows, subject to stochastic variation and credential availability.	Partially supported. QAOA fixed-angle parameters and sampled outputs are retained; selected VQE metadata reruns now retain optimized ansatz vectors for seed 74018 and for the final-budget selected seeds 74018, 74007, and 74001. Those metadata reruns are provenance artifacts, not replacements for the historical sampled VQE distributions.
Full campaign replay	Every stochastic campaign can be reproduced byte-for-byte from retained optimizer states, simulator logs, hardware calibration, and submitted-circuit metadata.	Not claimed. Some optimizer, MPS truncation, and submitted-job hardware metadata were not retained.

Table 3: Checksums for retained archives and primary cached summaries.

Artifact	SHA-256 checksum
case-studies/ds-mfg-qubo-qiskitopt/Fw_DSsmfgcasequboinformation.zip	aa20d793fc45c25c2bd7cc0a231877f71b225bf80daee1531cef6eb6cac5af2
ds_mfg_ibm_qaoa_pilot_fez_4096x3.zip	88af6f74b0fc10dcece69e783c37782649d1082e3220a88c61f389274665dff9
ds_mfg_direct_full_qubo_hardware_pilot.zip	fe7a60bd6576e66da9059d8aa6831424ef6fcbbf47bd934e60850b7933558091
case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_qaoa_juliquao_transfer_highread/qaoa_juliquao_transfer_summary.csv	10bb3bc9b746fb12a87eadc1fd2b819c1a3cacbc84025c9bd54797c1088c34b7
case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_vqe_reduced_flow_objective_final/vqe_reduced_top50_sampling_summary.csv	aac0299d49314b8d0e4d54b088f50cc728f1d212540613348ab62d209a132ee1
case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_gurobi_provenance/ip_exact_enumeration_summary.json	1298dcb07aa82c0adc05ead3fb1b3f4b86ae9ef38e9a276b98f97ec5a7606e5e
case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_gurobi_provenance/gurobi_local_pool_rerun_summary.json	13f7683c41cbf14b4225326e41778c6f42ae3979e4d1303af65c7132fe35a6e7
case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_vqe_reduced_flow_objective_selected_metadata/vqe_reduced_top50_sampling_summary.csv	df82de49b921e3b70d5d24f0896e77b6da64477c3a13cfc0a0761d6e9ebd2db
case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_hit_rate_reports/time_to_solution_report.csv	c27e70684f3b753efc4eb8cf2b75e79afe4b15402bcd6b3ca7d7a598310f2efd

VALIDATION FACTS

Table 4: Numerical validation facts used to support the manuscript claims.

Check	Retained value	Anchor
Exact repaired optimum	$F(y^*) = 11.7095$ at flow 1001110100111100011, ordered as $(f_{00}, \dots, f_{18})$.	ds_mfg_reduced_flow_objective/reduced_flow_summary.csv; ds_mfg_reduced_flow_objective/reduced_exact_top_flows.csv.
Gurobi pool bridge	All 36 stored Gurobi-pool flows match the repaired QUBO score within 3.1×10^{-13} .	MANUSCRIPT_FINDINGS.tex; Fw_DSsmfgcasequboinformation.zip.
Original Gurobi export metadata	Model capex: 19 binary variables, 20 constraints, 58 nonzeros; PoolSearchMode=2, PoolSolutions=36, solve time 0.073 s, 36 solutions found. Solver version, status code, MIPGap, PoolGap, and pool completeness/diversity intent were not retained in the original export.	Fw_DSsmfgcasequboinformation.zip/gurobi_pool_solutions_n36_20260126_105706.json.
Local Gurobi rerun validation	Gurobi 13.0.2 rerun of the same IP with PoolSearchMode=2 and PoolSolutions=100: OPTIMAL, MIPGap=0.0, best objective 11.7095, objective bound 11.7095, 36 solutions, local solve time 0.0077 s, and exact retained-pool agreement. PoolGap is null in the rerun summary and pool completeness/diversity is not claimed.	ds_mfg_gurobi_provenance/gurobi_local_pool_rerun_summary.json; ds_mfg_gurobi_provenance/gurobi_local_pool_rerun_solutions.csv.
Quadratic surrogate weakness	Global fit $R^2 \approx 0.999$, but the surrogate optimum repairs to 14.5505. The true repaired optimum is surrogate rank 33; the surrogate optimum is exact repaired rank 6.	ds_mfg_reduced_flow_objective/reduced_flow_summary.csv; Figure 1.
Low-energy surrogate audit	The exact top-50 repaired flows include 14 non-pool flows. In the lowest 1% repaired-flow set, exact-rank versus surrogate-rank Spearman correlation is 0.901. Exact enumeration wall time was not retained.	ds_mfg_reduced_flow_objective/reduced_exact_top_flows.csv; MANUSCRIPT_FINDINGS.tex.
Primary QAOA simulator check	JuliQAOA statevector probability on the reference optimum was 2.63×10^{-3} for the energy-targeted $p = 5$ angles. The cached Aer MPS sampled rate was 2.53×10^{-3} , inside its Wilson 95% interval $[2.35, 2.73] \times 10^{-3}$. MPS truncation logs were not retained.	ds_mfg_qaoa_juliquaoa_angle_search/juliquaoa_angle_summary.csv; ds_mfg_qaoa_juliquaoa_transfer_highread/qaoa_juliquaoa_transfer_summary.csv.

Table 5: Implications of missing provenance values for the manuscript claims.

Missing value	Affects main claim?	Affects full rerun?	Current mitigation
Original-export Gurobi final status and MIPGap	No for the current claim after rerun; the original export still lacks these fields, but the model/pool behavior has been rerun locally.	Partly; exact historical solver state is unavailable, but current solver provenance is retained.	The same flow is certified by exact enumeration over all 2^{19} flows and by a local Gurobi 13.0.2 rerun with OPTIMAL status, MIPGap=0.0, 36 solutions, and exact retained-pool agreement. PoolGap and original pool-generation intent remain unclaimed.
MPS truncation logs	No for the central QAOA sampling claim.	Yes, for simulator-forensics replay.	The energy-targeted QAOA row is cross-checked against independent JuliQAOA statevector probabilities, which agree with the Aer MPS sampled rate within the Wilson interval.
VQE optimized parameter vectors	No; VQE is secondary and reported as a retained sampled distribution.	Partly; selected fixed-circuit replay is now supported, but full historical optimizer replay is not claimed.	Metadata reruns retain optimized EfficientSU2 vectors for seed 74018 and for seeds 74018, 74007, and 74001 at the final sampling budget. The current metadata rerun produced 4 reference hits total and is provenance, not a replacement for the historical sampled VQE distributions reported in the manuscript.
Submitted hardware layout, depth, and two-qubit count	No, because the hardware row is a submission-and-scoring handoff only.	Yes, for hardware-performance diagnosis.	The manifest retains backend, job IDs, shots, seeds, logical circuit metadata, and scored counts. No noise-mechanism claim is made.
Hardware calibration or mitigation metadata	No under the current claim.	Yes, for calibrated hardware analysis.	The manuscript states that no readout or error mitigation is claimed and that the retained metadata cannot diagnose a specific noise source.
Exact enumeration wall time	No.	Partly, for cost accounting.	The enumeration is documented as exact over all 2^{19} flows; issue 24 tracks recording wall time if the enumeration script is rerun.

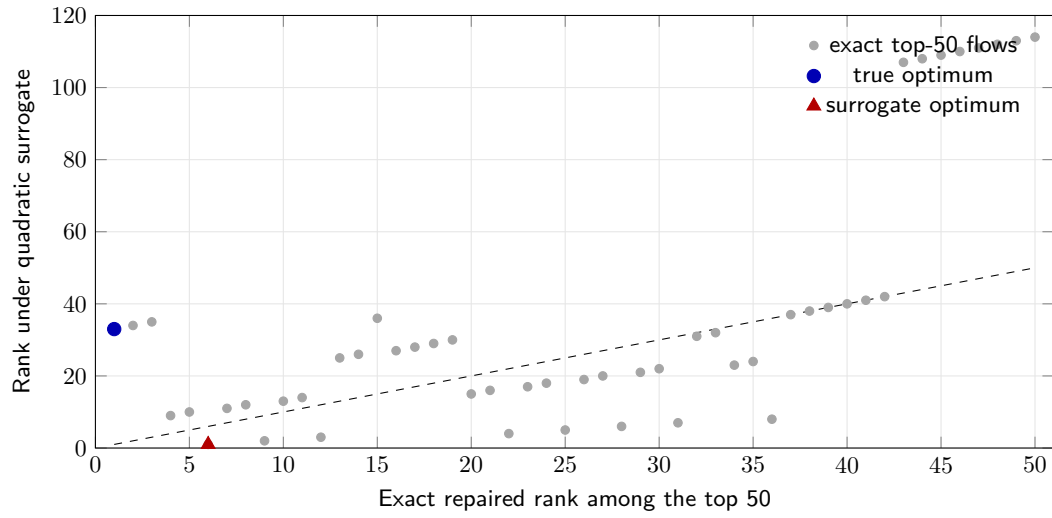


Figure 1. Surrogate rank audit for the exact top-50 repaired flows. The quadratic surrogate has high global R^2 but imperfect low-energy ordering: the exact optimum is only rank 33 under the surrogate, while the surrogate minimum repairs to the sixth-best exact flow. Quantum-sampling outcomes are therefore scored using the exact repaired objective F , not the surrogate $\hat{F}$.

VARIATIONAL AND HARDWARE PROVE- NANCE

Table 6: Variational-parameter provenance for the central QAOA and VQE rows.

Workflow	Circuit / ansatz	Objective and optimizer	Parameter source	Validation metric
Energy-targeted QAOA	$p = 5$ QAOA on the 19-flow surrogate.	Minimize $\mathbb{E}_{p_\theta}[\hat{F}(y)]$ with QiskitOpt.find_angles_bh; seed 91001; 5 basin-hopping iterations.	ds_mfg_qaoa_juliqaoa_angle_search/juliqaoa_angle_summary.csv; angles transferred beta-then-gamma to QiskitOpt.QAOA.	664 reference hits / 262144 reads; 39661 top-50 reads.
Top-10 QAOA diagnostic	$p = 5$ QAOA on the 19-flow surrogate.	Maximize probability on the enumerated exact top-10 set; oracle-aided because the repaired ranking is known.	Top-10 angle vector retained in ds_mfg_ibm_qaoa_pi_lot_fez_4096x3.zip/job_manifest.json; local 1007-hit diagnostic CSV was not retained.	1007 reference hits / 262144 reads in the retained manuscript summary; hardware handoff used this circuit.
Historical cached reduced-surrogate VQE	EfficientSU2; 152 initial parameters on the 19-flow surrogate.	COBYLA through QiskitOpt.VQE; 128 optimizer reads; 25 iterations; seeds 74001, 74007, and 74018 in the final follow-up.	Historical sampled distributions retained in ds_mfg_vqe_reduced_flow_objective_final/vqe_reduced_top50_sampling_summary.csv; historical optimized ansatz vectors were not retained.	Seed 74018 had 20 reference hits / 524288 reads, 152 top-10 reads, and 391 top-50 reads. The same historical three-seed final-follow-up artifact sums to 28 reference hits / 1572864 reads.
Current VQE meta-data/provenance rerun	EfficientSU2; 152 initial parameters on the 19-flow surrogate.	COBYLA through QiskitOpt.VQE; 128 optimizer reads; 25 iterations; seeds 74001, 74007, and 74018 at the final sampling budget.	Current metadata rerun persists optimized ansatz vectors in ds_mfg_vqe_reduced_flow_objective_selected_metadata/.	4 reference hits / 1572864 reads; used for provenance only and not to replace the historical VQE bar in the main manuscript figure or primary-outcomes table.

Table 7: Hardware pilot provenance retained in the submitted-job archive.

Item	Retained metadata	Limitation
Backend and jobs	ibm_fez; backend queried 2026-06-17 20:19 UTC; 156 qubits; 9 completed jobs; 4096 shots per job; repeats 1–3; transpile seeds 92001–92003; job IDs retained.	Queue time, calibration snapshot, readout mitigation, and error mitigation metadata were not retained.
Logical circuit	19 qubits and 19 classical bits; pre-transpile depth 86; operation counts: 19 h, 95 rx, 95 rz, 255 rzz, and 19 measurements; scoring reverses Qiskit count keys to $x_1, \dots, x_{19}$.	Submitted-job transpiled depth, two-qubit-gate count, physical layout, and measurement map were not retained.
Hardware outcome	36864 total reads; 6 top-50 reads; 1 top-10 read; 4 stored Gurobi-pool reads; 0 reference-optimum reads; best repaired flow rank 2 with objective 11.8105.	The pilot validates submission and scoring only; it cannot diagnose a specific noise mechanism.